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E-Commerce Sales & Customer Analysis Using SQL



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what are ..
Problem Statement

Content

- A growing e-commerce retail company wants to understand its sales performance, customer behavior, and order fulfillment health.

- The goal is to answer key business questions using SQL only no BI tools, no Python to demonstrate that meaningful business insight can be extracted directly from raw relational data.

Dataset Overview

customers table

800 rows (CustomerID, FullName, Email, SignUpDate, City, Country)

orders table

1,000 rows (OrderID, CustomerID, OrderDate, OrderStatus, TotalAmount)

products table

34 rows (ProductID, ProductName, Category, Brand, UnitPrice, OrderID)

Data period

January 2022 – mid 2026

Tools & SQL Concepts Used

Aggregate functions SUM, COUNT, AVG

GROUP BY and ORDER BY for summarization

Subqueries for percentage calculations

JOIN across multiple tables (customers ↔ orders ↔ products)

LEFT JOIN to find missing relationships (customers with no orders)

Date functions DATE_FORMAT, YEAR() for trend analysis

```
MariaDB [project]> # Is revenue growing or shinking month over month?
MariaDB [project]>  select
    -> date_format(OrderDate, '%Y-%m') as OrderMonth,
    -> count(OrderId) as TotalOrders,
    -> sum(TotalAmount) as TotalRevenue
    -> from orders
    -> group by OrderMonth
    -> order by OrderMonth;
```

OrderMonth	TotalOrders	TotalRevenue
2023-01	29	651760.35
2023-02	14	399141.55
2023-03	19	392613.03
2023-04	24	623657.18
2023-05	25	543655.42
2023-06	18	434723.31
2023-07	32	880385.34
2023-08	21	517066.59
2023-09	23	680943.10
2023-10	30	945005.15
2023-11	11	308136.06
2023-12	17	448549.20
2024-01	22	521437.94
2024-02	27	675730.82
2024-03	33	761049.03
2024-04	25	733502.51
2024-05	22	492837.73
2024-06	22	672737.24
2024-07	32	982505.01
2024-08	28	750507.07
2024-09	24	784215.23
2024-10	22	538829.89
2024-11	19	591957.84
2024-12	19	397173.97
2025-01	24	549486.02
2025-02	12	287965.39
2025-03	18	389755.95
2025-04	29	795135.46
2025-05	18	540165.08
2025-06	20	584753.29
2025-07	27	710771.69
2025-08	23	672536.57
2025-09	24	566788.50
2025-10	21	561903.05
2025-11	19	651183.86
2025-12	26	656278.77
2026-01	21	661195.22
2026-02	13	501352.81
2026-03	27	648031.26
2026-04	18	468290.93
2026-05	28	803871.04
2026-06	21	565010.22
2026-07	28	795828.33

Monthly Revenue Trend (Q1)

Revenue ranges ₹288K–₹982K/month, fairly volatile, no strong seasonal pattern

- Revenue swings between a low of ~₹288K (Feb 2025) and a high of ~₹982K (Jul 2024) per month
- Order volume also varies month to month, from 11 orders (Nov 2023) to 33 orders (Mar 2024)
- No clear repeating seasonal peak (e.g., no consistent festive/holiday spike) — the volatility looks closer to random fluctuation than a seasonal cycle
- Revenue and order count generally move together, meaning revenue swings are mostly driven by order volume, not by big changes in order size

Revenue by Order Status (Q2)

```
MariaDB [project]> # How much revenue is stuck in Pending/Cancelled vs actually Delivered?
MariaDB [project]> select
    -> OrderStatus,
    -> count(OrderID) as OrderCount,
    -> sum(TotalAmount) as Revenue,
    -> round(sum(TotalAmount)*100.0/(select sum(TotalAmount) from orders), 2) as PctOfTotalRevenue
    -> from orders
    -> group by OrderStatus
    -> order by Revenue desc;
+-------------+------------+------------+--------------------+
| OrderStatus | OrderCount | Revenue    | PctOfTotalRevenue |
+-------------+------------+------------+--------------------+
| Shipped     |        209 | 5700723.68 |              21.26 |
| Delivered   |        198 | 5580484.55 |              20.81 |
| Pending     |        205 | 5516532.64 |              20.57 |
| Processing  |        194 | 5117368.22 |              19.08 |
| Cancelled   |        194 | 4899566.81 |              18.27 |
+-------------+------------+------------+--------------------+
5 rows in set (0.022 sec)
```

- Nearly even split across 5 statuses (~18–21% each); ~37% tied up in Pending + Cancelled
- Roughly 37% of revenue (Pending + Cancelled combined) is either stuck in the pipeline or lost entirely

- Revenue is almost evenly distributed across 5 statuses: Shipped (21.3%), Delivered (20.8%), Pending (20.6%), Processing (19.1%), Cancelled (18.3%)
- Only ~21% of total revenue (Delivered) is fully realized and closed out

Top 10 Customers by Spend (Q3)

```
MariaDB [project]> # Who are the highest-value customers to prioritize for retention?
MariaDB [project]> select
    -> c.CustomerID,
    -> c.Name,
    -> c.City,
    -> c.Country,
    -> count(o.OrderID) as TotalOrders,
    -> sum(o.TotalAmount) as TotalSpend
    -> from customers c
    -> join orders o on c.CustomerID=o.CustomerID
    -> group by c.CustomerID, c.Name, c.City, c.Country
    -> order by TotalSpend desc
    -> limit 10;
+------------+--------------+-----------+---------+-------------+------------+
| CustomerID | Name         | City      | Country | TotalOrders | TotalSpend |
+------------+--------------+-----------+---------+-------------+------------+
| CUST-1377  | Sitara Singh | Delhi     | India   |           4 |  193596.54 |
| CUST-1514  | Reyansh Verma| Chennai   | India   |           5 |  172592.52 |
| CUST-1295  | Aarav Shah   | Jaipur    | India   |           4 |  163382.91 |
| CUST-1592  | Sitara Verma | Kolkata   | India   |           4 |  159015.85 |
| CUST-1152  | Riya Reddy   | Kolkata   | India   |           5 |  155889.66 |
| CUST-1530  | Ananya Kumar | Delhi     | India   |           4 |  152815.96 |
| CUST-1756  | Krishna Patel| Jaipur    | India   |           5 |  148314.21 |
| CUST-1134  | Krishna Shah | Kolkata   | India   |           4 |  148252.44 |
| CUST-1619  | Ananya Reddy | Delhi     | India   |           4 |  146737.81 |
| CUST-1244  | Riya Verma   | Hyderabad | India   |           4 |  144611.30 |
+------------+--------------+-----------+---------+-------------+------------+
10 rows in set (0.023 sec)
```

Highest-value customers spend ₹144K–₹193K each — small but valuable retention segment

- Top customer (Sitara Singh, Delhi) spent ₹193,596 across just 4 orders
- The top 10 customers spent between ₹144K–₹193K each, with 4–5 orders apiece
- These customers are spread across different cities (Delhi, Chennai, Jaipur, Kolkata, Hyderabad) — high value isn't concentrated in one region

Revenue by City (Q4)

```
MariaDB [project]> #Which regions generate the most revenue?
MariaDB [project]> select
    -> c.Country,
    -> c.City,
    -> count(o.OrderID) as TotalOrders,
    -> sum(o.TotalAmount) as TotalRevenue
    -> from customers c
    -> join orders o on c.CustomerID=o.CustomerID
    -> group by c.City
    -> order by TotalRevenue desc;
+---------+-----------+-------------+--------------+
| Country | City      | TotalOrders | TotalRevenue |
+---------+-----------+-------------+--------------+
| India   | Kolkata   |         113 |   3145905.74 |
| India   | Delhi     |         104 |   3020297.46 |
| India   | Pune      |         116 |   2857819.71 |
| India   | Surat     |         106 |   2775789.27 |
| India   | Chennai   |         101 |   2768455.05 |
| India   | Jaipur    |         105 |   2740649.93 |
| India   | Hyderabad |          88 |   2515270.42 |
| India   | Mumbai    |          95 |   2499922.25 |
| India   | Bangalore |          90 |   2368801.97 |
| India   | Ahmedabad |          82 |   2121764.11 |
+---------+-----------+-------------+--------------+
10 rows in set (0.011 sec)
```

Kolkata, Delhi, Pune lead; revenue is spread fairly evenly across 10 cities

- Kolkata leads (₹3.14M, 113 orders), followed by Delhi (₹3.02M) and Pune (₹2.86M)
- Ahmedabad is lowest of the 10 cities (₹2.12M) — still a substantial base
- The gap between the highest and lowest city is relatively small (~33%), showing revenue is fairly evenly distributed rather than concentrated in 1–2 metros

Average Order Value Trend (Q5)

```
MariaDB [project]>  # Is the average customer spending more or less per order over time?
MariaDB [project]> select
    -> date_format(OrderDate, '%Y-%m') as OrderMonth,
    -> round(avg(TotalAmount),2) as AvgOrderValue,
    -> count(OrderID) as TotalOrders
    -> from orders
    -> group by OrderMonth
    -> order by OrderMonth;
+------------+---------------+-------------+
| OrderMonth | AvgOrderValue | TotalOrders |
+------------+---------------+-------------+
| 2023-01    |      22474.49 |          29 |
| 2023-02    |      28510.11 |          14 |
| 2023-03    |      20663.84 |          19 |
| 2023-04    |      25985.72 |          24 |
| 2023-05    |      21746.22 |          25 |
| 2023-06    |      24151.30 |          18 |
| 2023-07    |      27512.04 |          32 |
| 2023-08    |      24622.22 |          21 |
| 2023-09    |      29606.22 |          23 |
| 2023-10    |      31500.17 |          30 |
| 2023-11    |      28012.37 |          11 |
| 2023-12    |      26385.25 |          17 |
| 2024-01    |      23701.72 |          22 |
| 2024-02    |      25027.07 |          27 |
| 2024-03    |      23062.09 |          33 |
| 2024-04    |      29340.10 |          25 |
| 2024-05    |      22401.71 |          22 |
| 2024-06    |      30578.97 |          22 |
| 2024-07    |      30703.28 |          32 |
| 2024-08    |      26803.82 |          28 |
| 2024-09    |      32675.63 |          24 |
| 2024-10    |      24492.27 |          22 |
| 2024-11    |      31155.68 |          19 |
| 2024-12    |      20903.89 |          19 |
| 2025-01    |      22895.25 |          24 |
| 2025-02    |      23997.12 |          12 |
| 2025-03    |      21653.11 |          18 |
| 2025-04    |      27418.46 |          29 |
| 2025-05    |      30009.17 |          18 |
| 2025-06    |      29237.66 |          20 |
| 2025-07    |      26324.88 |          27 |
| 2025-08    |      29240.72 |          23 |
| 2025-09    |      23616.19 |          24 |
```

AOV fluctuates between ₹20K–₹32K monthly, no consistent up/down trend

- AOV ranges from ~₹20.6K (Mar 2023) to ~₹32.7K (Sep 2024)
- No sustained upward or downward trend — AOV moves up and down month to month without a clear direction
- This is a different signal from the revenue trend (Slide 5): total revenue is driven more by how many orders happen than by customers spending more per order

Year-over-Year Growth (Q6)

```
MariaDB [project]> # How has the business grown year over year?
MariaDB [project]> select year(OrderDate) as OrderYear,
    -> count(OrderID) as TotalOrders,
    -> sum(TotalAmount) as TotalRevenue
    -> from orders
    -> group by OrderYear
    -> order by OrderYear;
+-----------+-------------+--------------+
| OrderYear | TotalOrders | TotalRevenue |
+-----------+-------------+--------------+
|      2023 |         263 |   6825636.28 |
|      2024 |         295 |   7902484.28 |
|      2025 |         261 |   6966723.63 |
|      2026 |         181 |   5119831.70 |
+-----------+-------------+--------------+
4 rows in set (0.007 sec)
```

- 2024 was peak (₹7.9M, 295 orders); 2026 (partial year) already at ₹5.1M
- 2026 (partial year, through mid-year data): ₹5.12M, 181 orders already — on pace to potentially match or exceed prior years

- 2023: ₹6.83M revenue, 263 orders
- 2024: ₹7.90M revenue, 295 orders — the peak year
- 2025: ₹6.97M revenue, 261 orders — a dip from 2024

Customers With Zero Orders (Q7)

```
MariaDB [project]> # Which signed-up customers have not made a single purchase yet?
MariaDB [project]> select c.CustomerID, c.Name, c.Email, c.SignUpDate from customers c left join orders o on c.CustomerId=o.CustomerID where o.OrderID is null;
+------------+----------------+--------------------------+------------+
| CustomerID | Name           | Email                    | SignUpDate |
+------------+----------------+--------------------------+------------+
| CUST-1004  | Arjun Singh    | customer.1004@example.com | 2022-09-28 |
| CUST-1005  | Pari Kumar     | customer.1005@example.com | 2025-08-20 |
| CUST-1009  | Sitara Kumar   | customer.1009@example.com | 2025-08-15 |
| CUST-1011  | Vihaan Singh   | customer.1011@example.com | 2022-01-10 |
| CUST-1013  | Ayaan Singh    | customer.1013@example.com | 2026-07-18 |
| CUST-1014  | Arjun Verma    | customer.1014@example.com | 2023-11-05 |
| CUST-1016  | Riya Shah      | customer.1016@example.com | 2023-07-27 |
| CUST-1019  | Ananya Verma   | customer.1019@example.com | 2026-06-29 |
| CUST-1021  | Riya Singh     | customer.1021@example.com | 2022-05-24 |
| CUST-1030  | Kiara Patel    | customer.1030@example.com | 2024-03-01 |
| CUST-1031  | Vihaan Patel   | customer.1031@example.com | 2025-06-05 |
| CUST-1033  | Ananya Jain    | customer.1033@example.com | 2026-06-07 |
| CUST-1036  | Diya Reddy     | customer.1036@example.com | 2022-02-28 |
| CUST-1038  | Ishaan Verma   | customer.1038@example.com | 2023-08-11 |
| CUST-1042  | Aditya Singh   | customer.1042@example.com | 2022-10-16 |
| CUST-1046  | Ananya Gupta   | customer.1046@example.com | 2026-02-08 |
| CUST-1047  | Aarav Mehta    | customer.1047@example.com | 2025-07-20 |
| CUST-1049  | Kiara Reddy    | customer.1049@example.com | 2024-04-03 |
| CUST-1057  | Krishna Verma  | customer.1057@example.com | 2022-06-02 |
| CUST-1065  | Saanvi Mehta   | customer.1065@example.com | 2025-02-09 |
| CUST-1074  | Sai Reddy      | customer.1074@example.com | 2024-03-20 |
| CUST-1082  | Aadya Verma    | customer.1082@example.com | 2024-09-08 |
| CUST-1083  | Sai Verma      | customer.1083@example.com | 2026-03-18 |
| CUST-1085  | Kiara Reddy    | customer.1085@example.com | 2022-01-14 |
| CUST-1090  | Krishna Jain   | customer.1090@example.com | 2023-09-05 |
| CUST-1095  | Arjun Verma    | customer.1095@example.com | 2023-02-27 |
| CUST-1100  | Aadya Patel    | customer.1100@example.com | 2024-05-27 |
| CUST-1105  | Aditya Sharma  | customer.1105@example.com | 2024-08-05 |
| CUST-1106  | Aadya Shah     | customer.1106@example.com | 2023-01-03 |
| CUST-1111  | Arjun Reddy    | customer.1111@example.com | 2025-08-09 |
| CUST-1117  | Kiara Singh    | customer.1117@example.com | 2024-05-05 |
| CUST-1118  | Sitara Jain    | customer.1118@example.com | 2022-11-26 |
| CUST-1119  | Aadya Sharma   | customer.1119@example.com | 2022-07-05 |
| CUST-1123  | Krishna Gupta  | customer.1123@example.com | 2025-04-17 |
| CUST-1138  | Ishaan Kumar   | customer.1138@example.com | 2024-01-04 |
| CUST-1140  | Aditya Sharma  | customer.1140@example.com | 2024-04-06 |
| CUST-1144  | Aadya Verma    | customer.1144@example.com | 2026-05-21 |
| CUST-1151  | Saanvi Reddy   | customer.1151@example.com | 2026-01-09 |
| CUST-1153  | Ayaan Jain     | customer.1153@example.com | 2023-06-22 |
| CUST-1156  | Vivaan Patel   | customer.1156@example.com | 2024-10-05 |
| CUST-1157  | Diya Reddy     | customer.1157@example.com | 2025-05-22 |
| CUST-1163  | Kiara Jain     | customer.1163@example.com | 2024-07-28 |
| CUST-1164  | Ishaan Jain    | customer.1164@example.com | 2024-08-20 |
| CUST-1167  | Riya Patel     | customer.1167@example.com | 2025-01-03 |
| CUST-1170  | Kiara Gupta    | customer.1170@example.com | 2025-07-13 |
| CUST-1171  | Arjun Shah     | customer.1171@example.com | 2025-11-25 |
| CUST-1184  | Riya Verma     | customer.1184@example.com | 2023-10-25 |
| CUST-1201  | Sitara Patel   | customer.1201@example.com | 2023-11-26 |
| CUST-1209  | Arjun Mehta    | customer.1209@example.com | 2024-12-01 |
```

Notable number of signed-up customers never purchased conversion opportunity

- A meaningful subset of the 800 signed-up customers have never placed a single order (sample shows dozens, e.g., CUST-1004, CUST-1005, CUST-1009, and more, spanning signup dates from 2022 to 2026)
- This isn't limited to recent signups — customers who joined as early as 2022 still show zero orders, so it's not simply "too new to have ordered yet"

Key Recommendations

Conclusion

- Recover stalled revenue: ~37% of revenue is tied up in Pending/Cancelled orders — investigate fulfillment bottlenecks and cancellation reasons.
- Retention focus: Top 10 customers each contribute ₹144K–₹193K — build a loyalty/VIP program around this segment.
- Re-engagement campaign: A meaningful number of signed-up customers have never ordered — target them with onboarding offers or first-purchase discounts.
- This project shows that core business questions — growth, customer value, fulfillment health, and conversion gaps — can be fully answered using SQL alone, without any additional BI or visualization tooling.



THANK YOU

